

Unit 4

Experimental Design and Sample Size

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Experimental design, controls, randomisation, replication, confounding, and planning sample size before data collection.
Central habit	Do not use statistics to repair a weak experiment. Design the experiment so the intended claim can be supported.

Contents

1 Purpose of Unit 4	2
2 The design-to-claim chain	2
3 Research questions and operational variables	3
4 Controls, blanks, standards, and baselines	4
5 Replication and independence	4
6 Randomisation	5
7 Blocking	5
8 Confounding variables	5
9 Error types and power	6
10 The smallest meaningful effect	7
11 Sample size for comparing two means	7
12 Worked example: plant height	8
13 Sample size for estimating a mean	9
14 Categorical data and expected counts	9
15 Planning for losses and failed samples	10
16 Practical versus statistical significance	11
17 IB-style design evaluation	11
18 Design checklist before data collection	12
19 Practice cases	12
20 Unit summary	13

1 Purpose of Unit 4

Units 1–3 built the foundation: measurement, descriptive statistics, visualisation, and distributions. Unit 4 moves one step earlier in the scientific process:

Before we ask which statistical test to use, we must ask whether the experiment can support the claim at all.

Statistical analysis is not a rescue procedure. If the design is confounded, the sample is not independent, the control is missing, or the sample size is too small for the intended effect, then a polished p-value cannot repair the argument.

Core thesis

The strength of a statistical conclusion is limited by the strength of the experimental design that produced the data.

By the end of this unit, students should be able to:

1. convert a broad scientific interest into a measurable research question;
2. identify independent, dependent, and controlled variables operationally;
3. distinguish controls, blanks, standards, baselines, and comparison groups;
4. distinguish biological replicates from technical replicates;
5. explain why randomisation and blocking improve experimental validity;
6. identify common confounding variables in biology and chemistry investigations;
7. explain Type I error, Type II error, and statistical power;
8. estimate sample size for common classroom designs;
9. justify sample size limitations honestly in IB-style writing.

2 The design-to-claim chain

A scientific claim is not produced directly by raw data. It is produced by a chain of decisions.



The narrowest point is often not calculation. It is the transition from an experimental idea to a design that produces interpretable evidence.

Bad design cannot be solved later

If all treated plants were placed near the window and all control plants were placed in the shade, a later t-test cannot tell whether the difference was caused by fertilizer or light. The design has already mixed two explanations together.

Plant growth example

Weak conclusion: “The fertilizer worked because treated plants were taller.”

Stronger conclusion: “The fertilizer group had a higher mean height than the control group under the tested conditions. The conclusion is limited by sample size, possible light-position effects, and the short growth period.”

3 Research questions and operational variables

A research question must be narrow enough that the variables can be measured.

Weak question	Problem	Improved question
Does fertilizer affect plants?	“Affect” is vague; plant response is unspecified.	How does nitrate fertilizer concentration affect mean height of wheat seedlings after 14 days?
Does temperature affect enzymes?	Enzyme, substrate, temperature range, and response are missing.	How does temperature from 20 °C to 60 °C affect catalase reaction rate measured by oxygen production per minute?
Does acid affect corrosion?	Acid type, concentration, metal, and outcome are not defined.	How does hydrochloric acid concentration affect mass loss of iron nails after 48 hours?

Operational definition means defining a variable by how it will actually be manipulated or measured.

Variable	Weak definition	Operational definition
Plant growth	“How much the plant grew”	Height from soil surface to tallest leaf tip, measured in cm after 14 days.
Reaction rate	“How fast the reaction is”	Volume of oxygen gas produced per minute in mL min ⁻¹ .
pH condition	“Acidic environment”	Buffer solution adjusted to pH 4.00, 5.00, 6.00, and 7.00 using a calibrated pH probe.

Quick check

Rewrite the following as operational research questions:

1. Does salt affect germination?
2. Does temperature affect vitamin C?
3. Do different disinfectants kill bacteria?

For each one, specify the independent variable, dependent variable, units, and time scale.

4 Controls, blanks, standards, and baselines

Students often use the word “control” too loosely. In experimental work, different reference conditions have different jobs.

Reference type	Purpose	Example
Negative control	Shows what happens when the active cause is absent.	Enzyme assay without enzyme, or bacteria without antibiotic.
Positive control	Shows that the method can detect a known effect.	Known glucose solution for a glucose test strip.
Blank	Corrects for background signal.	Colorimeter cuvette containing solvent and reagents but no analyte.
Standard	Provides known values for calibration.	Copper sulfate solutions of known concentration for absorbance calibration.
Baseline	Starting value or untreated reference condition.	Initial plant height before fertilizer treatment.

Control logic

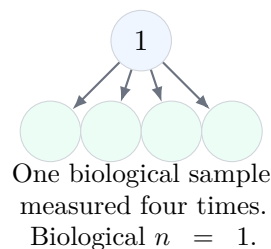
A control is not merely an extra group. It is an argument against a specific alternative explanation.

Example: if a colorimeter reading changes after adding a reagent, a blank helps decide whether the signal comes from the analyte or from the reagent/background absorbance.

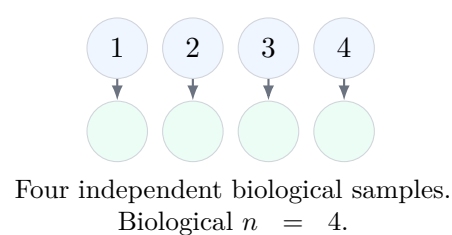
5 Replication and independence

Replication is not just “doing it many times.” The repeats must match the claim.

Technical repeats only



Biological replicates



Technical replicates estimate measurement precision. **Biological replicates** estimate biological variation and support inference beyond one sample.

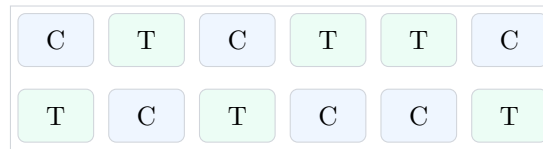
Common IA error

A student measures the same bacterial culture five times and writes $n = 5$. If all measurements come from the same culture, the biological sample size is still $n = 1$. The five readings may improve measurement precision, but they do not represent five independent cultures.

6 Randomisation

Randomisation means using a random procedure to allocate samples, treatments, or measurement order. It protects the experiment against hidden patterns.

Randomly assigned pots across one bench



Randomisation does not guarantee perfect balance in a small experiment, but it prevents the researcher from systematically assigning stronger, healthier, larger, or more convenient samples to one condition.

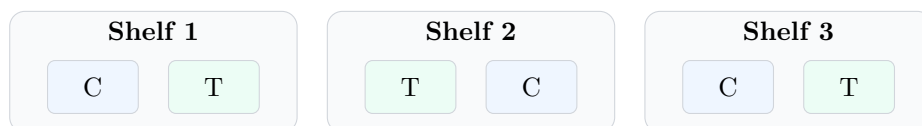
Measurement order

If all control titrations are performed first and all treatment titrations are performed after lunch, then time, fatigue, reagent drift, or temperature may be mixed with treatment. Randomising the order of measurements can reduce this risk.

7 Blocking

Blocking is used when there is a known source of variation that cannot be removed but can be organised.

Example: plant trays may sit on different shelves with slightly different light exposure. Instead of ignoring this, place both treatment and control inside each shelf block.

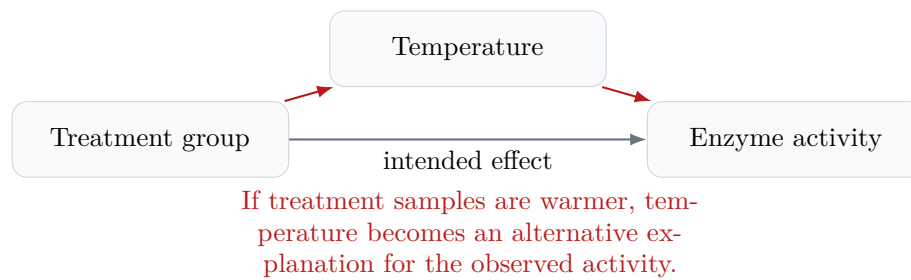


Each block contains both conditions. Shelf variation is no longer perfectly confused with treatment.

Blocking is especially useful in school labs because the room is rarely perfectly uniform. Benches, shelves, incubator positions, water bath corners, and time slots can all become blocks.

8 Confounding variables

A **confounder** is a variable that is mixed with the independent variable and also affects the dependent variable.



Investigation	Possible confounder	Design response
Fertilizer and plant height	Light exposure, initial seedling size, soil batch	Randomise positions; measure initial height; block by shelf or tray.
Temperature and enzyme rate	pH drift, substrate depletion, timing differences	Use buffers; prepare equal substrate concentrations; randomise measurement order.
Antibiotic and bacterial growth	Different inoculum density, plate dryness, incubation position	Standardise inoculum; rotate/incubate plates consistently; randomise plate positions.
Titration method comparison	Different operators, endpoint colour judgement, reagent ageing	Counterbalance operator; standardise endpoint rule; use fresh standardised solutions.

9 Error types and power

In hypothesis testing, two broad errors are possible.

	Reality: no real effect	Reality: real effect exists
Decision: reject H_0	Type I error: false positive. Controlled by α .	Correct detection. Probability is power.
Decision: fail to reject H_0	Correct non-detection.	Type II error: false negative. Probability is β .

Power is the probability that the experiment detects an effect of a specified size, assuming that effect really exists.

$$\text{Power} = 1 - \beta$$

A common planning target is 80% power. This means: if the chosen meaningful effect truly exists, the design would detect it about 80% of the time under repeated experiments.

Power is conditional

Power is not the probability that your hypothesis is true. It is the probability of detecting a specified effect under a specified model, sample size, error rate, and variability.

10 The smallest meaningful effect

Before calculating sample size, define the smallest effect worth detecting. This is not necessarily the smallest mathematically detectable difference.

Context	Tiny but detectable	Meaningful effect
Plant growth	A mean difference of 0.2 cm	A difference large enough to matter biologically or agriculturally.
Titration method	A difference of 0.01 cm ³	A difference larger than normal endpoint and burette uncertainty.
Enzyme activity	A 1% change in rate	A change large enough to affect the biological interpretation.

For comparing two means, a useful standardised effect size is:

$$d = \frac{\delta}{\sigma}$$

where δ is the mean difference you want to detect and σ is the expected standard deviation.

Planning rule

The smaller the meaningful effect, and the larger the natural variation, the larger the required sample size.

11 Sample size for comparing two means

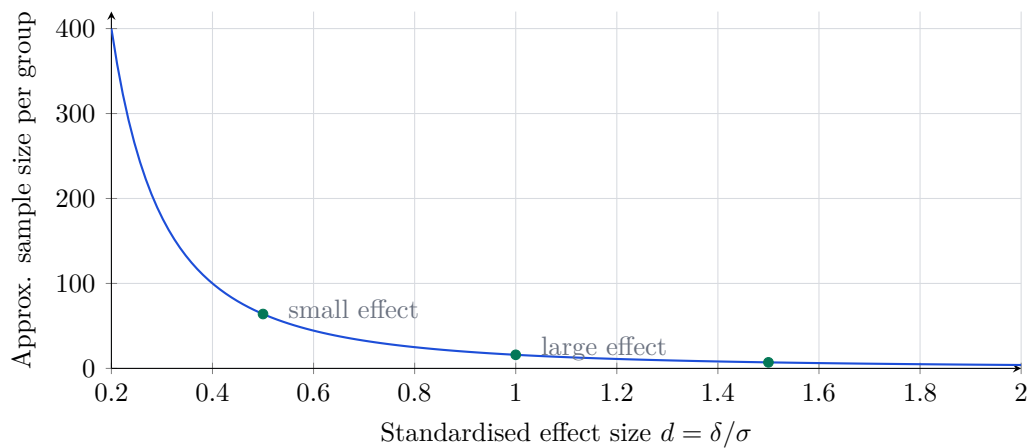
For two independent groups with approximately equal sizes, a rough planning formula is:

$$n \approx \frac{2(z_{\alpha/2} + z_{\beta})^2 \sigma^2}{\delta^2}$$

where n is the sample size per group, σ is the expected standard deviation, δ is the smallest meaningful mean difference, α is the Type I error rate, and power is $1 - \beta$.

For $\alpha = 0.05$ and 80% power, this is approximately:

$$n \approx \frac{16}{d^2}$$



The curve shows a brutal fact: detecting a small effect requires many more samples than detecting a large effect.

Four-times rule

If the meaningful effect becomes half as large while variability stays the same, the required sample size becomes about four times larger. This follows from the squared term δ^2 in the denominator.

12 Worked example: plant height

A student wants to test whether fertilizer increases plant height after 14 days.

Pilot data suggest:

$$\sigma \approx 5 \text{ cm}$$

The student decides that the smallest meaningful difference is:

$$\delta = 5 \text{ cm}$$

So:

$$d = \frac{\delta}{\sigma} = \frac{5}{5} = 1$$

Using the rough 80% power rule:

$$n \approx \frac{16}{1^2} = 16$$

The student should plan for about 16 independent plants per group, before allowing for losses.

If the student wants to detect only 2.5 cm:

$$d = \frac{2.5}{5} = 0.5$$

$$n \approx \frac{16}{0.5^2} = 64$$

IB limitation

A school investigation may not have enough time or material for ideal sample size. That does not make the work worthless. It means the conclusion must be written as limited, preliminary, or suggestive rather than definitive.

13 Sample size for estimating a mean

Sometimes the goal is not to detect a difference, but to estimate a value with acceptable precision. For estimating a mean with approximate 95% confidence, a rough formula is:

$$n \approx \left(\frac{z\sigma}{E} \right)^2$$

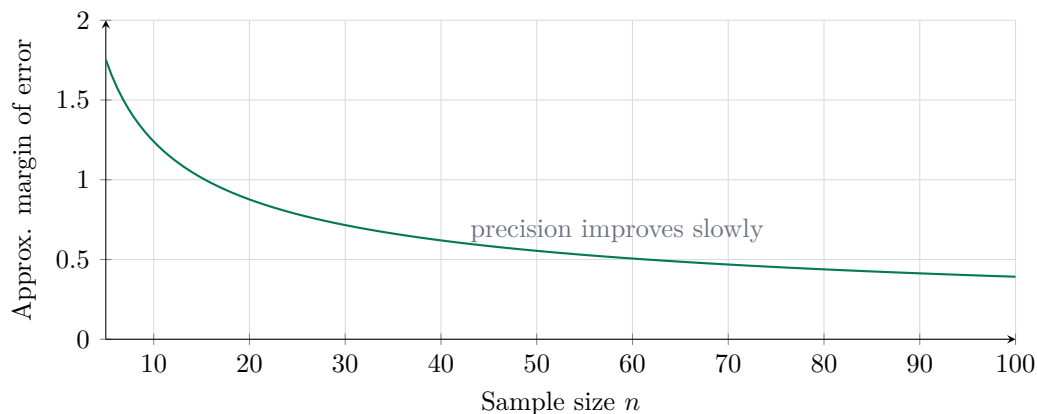
where E is the desired margin of error and $z \approx 1.96$ for 95% confidence.

Calcium concentration example

Suppose previous water samples suggest $\sigma \approx 2 \text{ mg L}^{-1}$. You want to estimate the mean calcium concentration with margin of error $E = 0.5 \text{ mg L}^{-1}$.

$$n \approx \left(\frac{1.96 \times 2}{0.5} \right)^2 \approx 61.5$$

So about 62 independent samples would be needed for that precision.



Precision improves with $\sqrt{n}$, not directly with n . Doubling sample size does not halve the margin of error.

14 Categorical data and expected counts

For categorical data, such as Mendelian offspring counts or survival counts, the sample size problem often appears as small expected counts.

For a chi-square goodness-of-fit test, a common classroom rule is:

Expected count in each category should usually be at least 5.

If the smallest expected proportion is $p_{\min}$, then a rough minimum is:

$$n \geq \frac{5}{p_{\min}}$$

Expected ratio	Smallest category	Calculation	Minimum n
3:1	$1/4 = 0.25$	$5/0.25$	20
9:3:3:1	$1/16 = 0.0625$	$5/0.0625$	80
1:1:1:1	$1/4 = 0.25$	$5/0.25$	20

Minimum is not enough

Meeting the expected-count rule only helps the chi-square approximation behave better. It does not guarantee high power or a strong biological conclusion.

For small categorical samples, exact binomial tests, Fisher's exact test, or simulation-based methods may be more appropriate than chi-square approximations.

15 Planning for losses and failed samples

Real experiments lose samples. Seeds fail to germinate, cultures contaminate, cuvettes are scratched, gas syringes leak, and titration trials are sometimes rejected for clear procedural reasons.

If n_{needed} usable samples are required and the expected loss rate is L , then:

$$n_{\text{start}} = \frac{n_{\text{needed}}}{1 - L}$$

Contamination allowance

If 30 usable bacterial plates are needed and about 15% may be lost:

$$n_{\text{start}} = \frac{30}{1 - 0.15} = 35.3$$

The student should prepare about 36 plates.

Likely loss source	Example	Planning response
Contamination	Microbial plates with unwanted colonies	Prepare extra plates; define exclusion criteria before analysis.
Instrument failure	Gas syringe leak, colorimeter error	Pilot apparatus; include spare measurements.
Biological non-response	Seeds not germinating	Start with extra seeds; report germination failure.
Procedural invalidity	Titration overshoot endpoint	Define repeat/reject criteria before final analysis.

16 Practical versus statistical significance

A result can be statistically significant but practically unimportant. A result can also look biologically large but fail to reach statistical significance because the sample is small.

Situation	Possible result	Better interpretation
Huge sample, tiny effect	$p < 0.001$, mean difference 0.1 cm	Statistically detectable, but possibly biologically trivial.
Small sample, large observed effect	$p = 0.12$, mean difference 5 cm	Suggestive pattern, but uncertainty is large.
Weak design, significant p-value	$p = 0.03$, but treatment confounded with light	Statistical result does not isolate the cause.

Reporting standard

A strong conclusion combines effect size, uncertainty, design validity, and the statistical result. A p-value alone is not enough.

17 IB-style design evaluation

Students often write vague evaluations such as “human error affected the results.” This is too weak. A useful evaluation identifies a mechanism and explains how it could influence the conclusion.

Weak evaluation	Stronger evaluation
Human error may have occurred.	Endpoint colour judgement could have varied between titrations, causing random error in final volume readings. A more precise endpoint method or the same trained observer should be used.
The sample size was small.	The small number of biological replicates reduces power and gives wide uncertainty around the estimated treatment effect. The result should therefore be treated as preliminary.
Temperature was not controlled.	Temperature affects enzyme rate directly. If treatment and control samples were measured at different times, temperature could confound the observed difference.
More repeats are needed.	More independent biological replicates are needed because technical repeats only estimate measurement precision, not biological variability.

A defensible IB-style limitation should include:

1. the specific source of uncertainty or bias;
2. the direction or kind of effect it may have caused;
3. whether it threatens reliability, validity, or generalisability;
4. a concrete improvement.

18 Design checklist before data collection

Before collecting data, students should be able to answer the following questions.

Pre-analysis design checklist

1. What exact claim do I want the data to support?
2. What is my independent variable, and how is it manipulated?
3. What is my dependent variable, and how is it measured?
4. What are the most dangerous confounders?
5. What control or comparison group is needed?
6. Are my replicates biologically independent?
7. Should I randomise sample positions or measurement order?
8. Is there a known source of variation that should be blocked?
9. What is the smallest meaningful effect?
10. Is my planned sample size large enough to make a useful claim?
11. What sample losses are likely, and how will I handle them?
12. What would make a result invalid before analysis begins?

19 Practice cases

Case 1: Enzyme temperature investigation

A student tests catalase activity at 20 °C, 30 °C, 40 °C, and 50 °C. The student performs all 20 °C trials first, then all 30 °C trials, and so on. The hydrogen peroxide bottle remains open throughout.

Identify:

1. one possible confounder;
2. one randomisation improvement;
3. one control or standardisation step;
4. what kind of conclusion would be too strong.

Case 2: Germination and salt

A student places 10 seeds in pure water and 10 seeds in salt solution. After 5 days, 8 seeds germinate in water and 4 germinate in salt solution.

Discuss:

1. whether the sample size is enough for a strong conclusion;
2. whether the outcome is categorical or continuous;
3. what effect size could be reported;
4. how the experiment could be expanded.

Case 3: Sample size planning

A pilot study suggests that plant height has standard deviation $\sigma = 4$ cm. A student wants to detect a mean difference of $\delta = 2$ cm between control and treatment.

1. Calculate $d = \delta/\sigma$.
2. Use $n \approx 16/d^2$ to estimate sample size per group.
3. Explain why this may be impractical in a short school investigation.
4. Write a cautious conclusion sentence suitable for a small-sample version of the study.

20 Unit summary

Main lesson

Experimental design is the foundation of statistical inference. A good design makes alternative explanations less plausible before any p-value is calculated.

Students should leave Unit 4 with these habits:

1. State the intended claim before choosing the statistical method.
2. Define variables operationally.
3. Use controls as arguments against alternative explanations.
4. Treat biological replication as separate from technical repetition.
5. Use randomisation and blocking to protect validity.
6. Search actively for confounders.
7. Plan sample size from meaningful effect size and expected variability.
8. Report limitations specifically, not with vague phrases.

A concise final sentence for students:

The experiment should be designed so that the data can answer the question; statistics then measures how strong and uncertain that answer is.